CURRICULUM VITAE

PERSONAL INFORMATION



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RESEARCH INTEREST

AI-driven drug discovery with a focus on agentic AI frameworks for predictive modeling and decision-making. Experience in benchmarking standard AI/ML methods across internal and external drug discovery datasets, and in designing AI agent workflows to improve transparency, robustness, and automation. Strong background in cheminformatics, computational chemistry, and machine learning in Python, including large language models (LLMs), graph neural networks, and chemical foundation models.

EDUCATION

Nov 2022 - Present	PhD in Computational Biology, Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), New Delhi, India - CGPA: 8.6/10
Aug 2018 - Jul 2020	M.Sc. in Bioinformatics, Central University of South Bihar (CUSB), Gaya, Bihar, India - CGPA 7.79
Aug 2014 - Jul 2017	B.Sc. (Hons.) in Zoology, Magadh University, Gaya, Bihar, India 61.6%

RESEARCH EXPERIENCE

Nov 2022 - Present — PhD Researcher, IIIT-Delhi, New Delhi, India

Conducting interdisciplinary research at the intersection of AI, computational biology, chemoinformatics, and precision oncology.

- Developed ChemicalDice, a deep learning-based multimodal molecular representation framework for integrating complementary chemical representations and generating unified molecular embeddings for drug discovery and molecular property prediction.
- Developed OncoGen.AI, an end-to-end platform for automated genomic analysis, knowledge-driven variant interpretation, and clinical reporting in precision oncology.

- Fine-tuned BioMistral-7B on the EvoAge aging knowledge graph (1.04B triples, 6 species) using GraphGen for synthetic QA generation and LLaMA-Factory for Supervised Fine-Tuning with LoRA, identifying a balanced configuration that minimized graph loss while avoiding degradation on MedMCQA, PubMedQA, etc. Evaluated using the LM Evaluation Harness.

Mar 2022 - Oct 2022 Data Scientist, INDX Technology, Mumbai, Maharashtra, India

- Developed machine learning models to predict PD-1 immunotherapy responders and non-responders using patient-level molecular and clinical data. Developed a patient-matching framework using clinical and molecular metadata to identify patients with similar characteristics for downstream analysis.
- Developed computational tools and interactive interfaces for biomedical data visualization and exploration, including three-dimensional sample visualization.

Mar 2021 - Feb 2022 Junior Data Analyst, INDX Technology, Mumbai, Maharashtra, India

- Processed and analyzed biomedical and clinical datasets to support downstream data-driven applications.
- Contributed to the development of interactive user interfaces for the visualization and exploration of biological and patient-level data.

Aug 2020 - Oct 2020 Research Associate, NIPGR, New Delhi, India

- Conducted computational characterization of genes in the Cicer arietinum genome. Analyzed genomic sequence information and gene features to investigate their potential biological functions.

Jan 2020 - Jul 2020 Research Intern, IIIT-Delhi, New Delhi, India

- Conducted integrative transcriptomic analysis of non-small cell lung cancer (NSCLC) using mRNA-seq and miRNA-seq datasets.
- Analyzed transcriptomic profiles to identify differentially expressed genes, molecular signatures, and regulatory patterns associated with NSCLC. Applied computational and statistical approaches for high-dimensional sequencing data analysis and biological interpretation

ACADEMIC RESPONSIBILITIES

Winter 2023	Teaching Assistant, *Biostatistics*
Monsoon 2023	Teaching Assistant, *Foundations of Modern Biology*
Monsoon 2024	Teaching Assistant, *Machine Learning*
Monsoon 2025	Teaching Assistant, *Computing for Medicine*
Winter 2026	Teaching Assistant, *Cellular Biophysics*

PUBLICATIONS

1. **Kumar, Suvendu***, Solanki, Saveena*, Gupta, Mudit*, Chauhan, Sonam*, Mohanty, Sanjay Kumar*, Satija, Shiva, Sharma, Abhinav Kumar, Duari, Subhadeep, Sharma, Arushi, Shome, Raidhani, Gautam, Vishakha, Arora, Sakshi, Ali, Syed Yasser, Raza, Adnan, Sinha, Sourav, Mittal, Aayushi, Sengupta, Debarka, Murugan, Natarajan Arul, and Ahuja, Gaurav. (2026). **Scalable Molecular Representations Enabled by Multimodal Fusion and Sequence Distillation.** *Nature Communications.* https://doi.org/10.1038/s41467-026-77700-z
2. Shome, Raidhani*, **Kumar, Suvendu***, Mohanty, Sanjay Kumar*, Dholakia, Dhwani*, Sharma, Arushi, Solanki, Saveena, Sharma, Sakshi, Chauhan, Sonam, Satija, Shiva, Diwan, Himanshi, Panigrahi, Manoj Kumar, Tayal, Juhi, Mehta, Anurag, and Ahuja, Gaurav. (2026). **OncoGen.AI: An Integrated Platform for Automated Genomic Analysis and Reporting in Precision Oncology.** *npj Precision Oncology.* https://doi.org/10.1038/s41698-026-01578-9
3. Satija, Shiva*, Mohanty, Sanjay Kumar*, Jorvekar, Sachin B.*, Verma, Anshul, Solanki, Saveena, Gautam, Vishakha, Arora, Sakshi, Mittal, Aayushi, Sharma, Arushi, Chauhan, Sonam, Duari, Subhadeep, **Kumar, Suvendu**, Rai, Avdhesh, Das, Anupam, Kakati, Kaberi, Das, Kishore, Sarma, Anupam, Tayal, Juhi, Sengupta, Debarka, Mehta, Anurag, Borkar, Roshan M., and Ahuja, Gaurav. (2026). **MutAIverse: An AI-Powered, Mechanism-Backed Platform for Discovering Novel DNA Adducts and Their Precursor Genotoxins.** *Journal of Cheminformatics.* https://doi.org/10.1186/s13321-026-01238-x
4. Mittal, Aayushi*, Gupta, Mudit*, Mohanty, Sanjay Kumar*, Gaur, Aakash*, Solanki,

Saveena*, Ali, Syed Yasser, Raza, Adnan, Sharma, Pranjal, **Kumar, Suvendu**, Gautam, Vishakha, Satija, Shiva, Katyayan, Nikhil, Tiwari, Kunal, Sharma, Anmol Kumar, Kaur, Dilraj, Sengupta, Debarka, Gupta, Shashi Kumar, and Ahuja, Gaurav. (2026). **Evolutionary-guided advanced deep-learning architecture powers mammalian GPCRome agonist predictions.** *Cell Reports.* https://doi.org/10.1016/j.celrep.2026.117003

5. Mohanty, Sanjay Kumar*, Mittal, Aayushi*, Farooqi, Namra*, Gaur, Aakash, Duari, Subhadeep, Solanki, Saveena, Sharma, Anmol Kumar, Arora, Sakshi, **Kumar, Suvendu**, Gautam, Vishakha, Dixit, Nilesh Kumar, Subramanian, Karthika, Ghosh, Tarini Shankar, Sengupta, Debarka, Gupta, Shashi Kumar, Murugan, Arul Natarajan, Sharma, Deepak, and Ahuja, Gaurav. (2025). ***Deep Learning Reveals Endogenous Sterols as Allosteric Modulators of the GPCR–Gα Interface.*** *eLife, 14, RP106397.* https://doi.org/10.7554/eLife.106397
6. Solanki, Saveena*, Mohanty, Sanjay Kumar*, Satija, Shiva, Chauhan, Sonam, Bandaru, N. V. M. R., Dukare, S., Tiwari, N. K., Kumar, N. R., Aravind, A. B., Mukherjee, S., Gautam, Vishakha, Arora, Sakshi, **Kumar, Suvendu**, Duari, Subhadeep, Sharma, Arushi, Shome, Raidhani, Sengupta, Debarka, Abbineni, C., Samajdar, S., and Ahuja, Gaurav. (2025). **SynGlue: AI-Driven Designer for Clinically Actionable Multi-Target Therapeutics.** *bioRxiv.* https://doi.org/10.1101/2025.08.28.672835
7. Gautam, Vishakha*, Duari, Subhadeep*, Solanki, Saveena, Gupta, Mudit, Mittal, Aayushi, Arora, Sakshi, Aggarwal, Anmol, Sharma, Anmol Kumar, Tyagi, Sarthak, Rathod, Kunal Pankajbhai, Sharma, Arushi, Chauhan, Sonam, Satija, Shiva, **Kumar, Suvendu**, Mohanty, Sanjay Kumar, Tayal, Juhi, Dixit, Nilesh Kumar, Sengupta, Debarka, Mehta, Anurag, and Ahuja, Gaurav. (2025). **scCamAge: A Context-Aware Prediction Engine for Cellular Age, Aging-Associated Bioactivities, and Morphometrics.** *Cell Reports.* https://doi.org/10.1016/j.celrep.2025.115270
8. Gupta, Anku*, Choudhary, Mohit*, Mohanty, Sanjay Kumar*, Mittal, Aayushi*, Gupta, Krishan*, Arya, Aditya*, **Kumar, Suvendu**, Katyayan, Nikhil, Dixit, Nilesh Kumar, Kalra, Siddhant, Goel, Manshi, Sahni, Megha, Singhal, Vrinda, Mishra, Tripti, Sengupta, Debarka, and Ahuja, Gaurav. (2021). **Machine-OIF-Action: A Unified Framework for Developing and Interpreting Machine-Learning Models for Chemosensory Research.** *Bioinformatics.* https://doi.org/10.1093/bioinformatics/btaa1104
9. Gupta, Krishan*, Mohanty, Sanjay Kumar*, Mittal, Aayushi*, Kalra, Siddhant*, **Kumar, Suvendu**, Mishra, Tripti, Ahuja, Jatin, Sengupta, Debarka, and Ahuja, Gaurav. (2021). **The Cellular Basis of Loss of Smell in 2019-nCoV-Infected Individuals.** *Briefings in Bioinformatics, 22(2), 873–881.* https://doi.org/10.1093/bib/bbaa168

CONFERENCES & SYMPOSIA

2024 *Post-transcriptional Regulation in Ageing and Age-Related Diseases* (EMBO)

2023 *Algorithms that Transformed Molecular Biology* (ATMB), IIIT-Delhi

2020 *Research Innovation and Incubation Showcase* (RIISE), IIIT-Delhi

ACADEMIC FELLOWSHIPS

Aug 2024 - May 2025 CHANAKYA Doctoral Fellowship – Comprehensive and Holistic Advancement of National Knowledge Yield and Analytics

TECHNICAL SKILLS

- **Machine Learning & AI:** Machine Learning, Deep Learning, Generative AI, Molecular Representation Learning, Molecular Property Prediction, De novo Drug Design
- **Cheminformatics:** QSAR/QSPR Modeling, Molecular Docking, Molecular Dynamics (MD) Simulations, RDKit, MOPAC, Open Babel, PyMOL, AutoDock Vina, LigBuilder
- **LLM Engineering & MLOps**: LLaMA-Factory, SGLang, GraphGen, vLLM, lm-eval, Synthetic QA Generation, Model Evaluation & Benchmarking
- **Computational Genomics & Transcriptomics:** NGS Analysis, Variant Calling, Mutational Profiling, Bulk RNA-seq, Single-cell RNA-seq (scRNA-seq), Differential Expression Analysis
- **Programming & Scientific Computing:** Python, R, Bash, Linux, Git, Docker, HPC
- **Machine Learning Frameworks:** PyTorch, TensorFlow, scikit-learn
- **Bioinformatics & Genomics:** GATK, SAMtools, FastQC, Gcoupler
- **Visualizations :** Adobe Illustrator, Origin

LANGUAGES

- **English:** Full professional proficiency
- **Hindi:** Full professional proficiency

REFERENCES

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